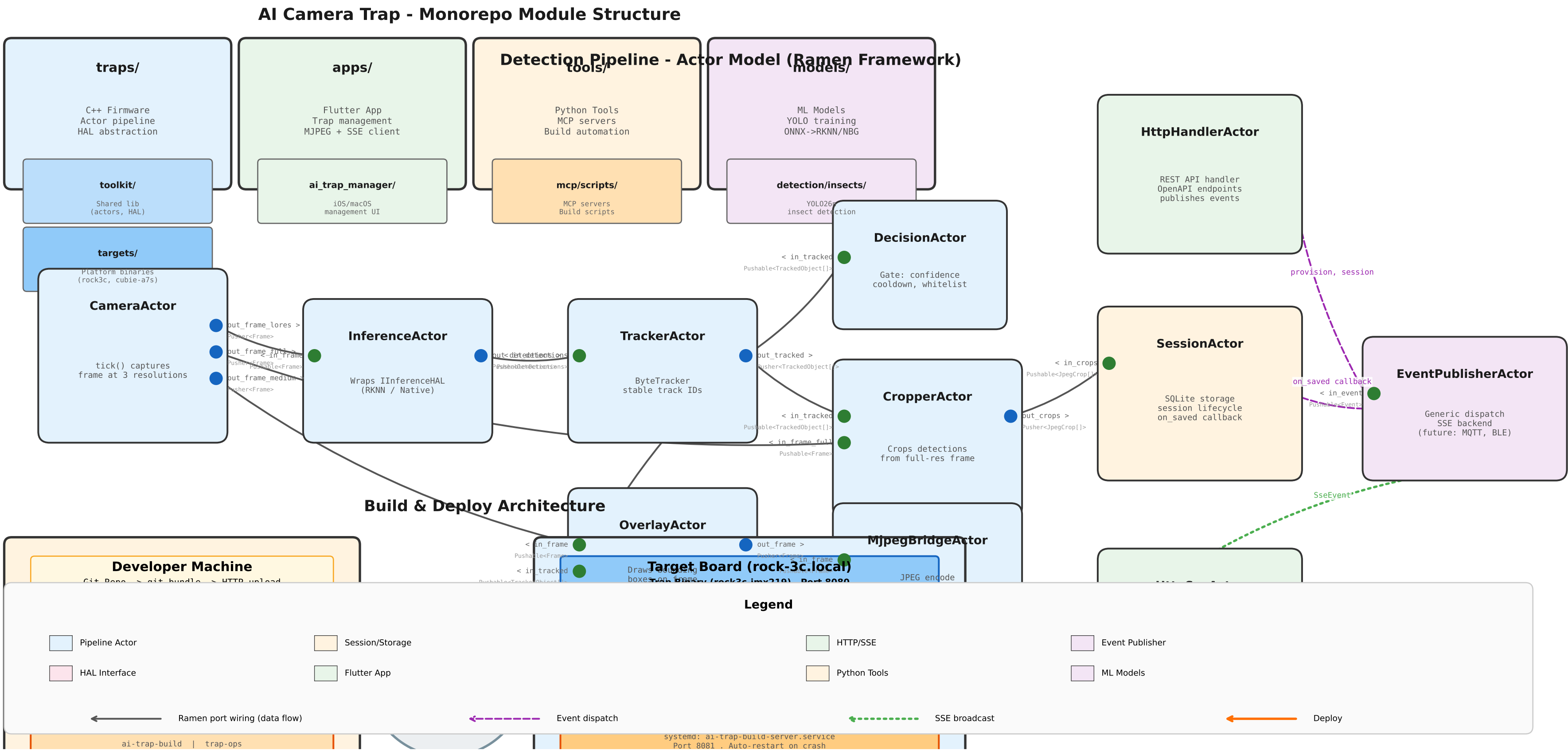




Camera (Hardware)

V4L2 + ISP
Frame Capture

YOLO Inference
(NPU: RKNN)

ByteTracker
Multi-Object
Tracking

Decision
Gate Logic

Cropper +
JPEG Encode

SQLite +
Disk Storage

Overlay +
MJPEG Stream

HTTP SSE
Event Stream

Flutter App
(MJPEG + SSE)

tracked objects

events

MJPEG stream

SSE events

Data Flow Overview